Pre-Analysis Plan (Revised): The Effect of Having Daughters on Attitudes and Behaviors

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Overview

This pre-analysis plan (PAP) outlines the empirical strategy and hypotheses we intend to test using the German Socio-Economic Panel (SOEP) data. We aim to investigate whether having daughters influences attitudes toward gender equality, political preferences, and intra-household dynamics.

Hypothesis 1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

Dependent Variable

To measure attitudes toward gender equality, we will use the following single item from the 2019 survey wave:

- `plh0395i05`: An ordinal measure capturing agreement with the statement, "Schools and kindergartens should encourage children to engage in activities that are not typical for their gender."

Main Independent Variable

To identify the effect of child gender, our analysis will be restricted to parents with only same-gender children. The primary independent variable is:

- `has_only_daughters`: A binary indicator equal to 1 for parents with only daughters, and 0 for parents with only sons.

Sample Restrictions

- Adults aged 18 and above.
- Respondents with at least one child.
- The analysis is restricted to the 2019 cross-section, as the dependent variable is only available in that wave.

Estimation Strategy

- We will estimate a single ordered probit model.
- The model will use robust standard errors.
- Controls will include: age (continuous), age squared, respondent gender (`sex`), education level in years (`pgbilzeit`), place of residence (East/West Germany, from `sampreg`), and a binary indicator for being born in Germany (derived from `corigin`).

Interpretation

We consider the hypothesis supported if the coefficient on `has_only_daughters` is positive and statistically significant at the 5% level.

Hypothesis 2: Having daughters leads to more left-leaning political preferences

Dependent Variable

- `plh0004`: An 11-point scale of political self-placement from left (0) to right (10), available in 2005, 2009, 2014, and 2019.

Main Independent Variable and Sample Restrictions

The main independent variable is `has_only_daughters`, and the sample is restricted to parents with only same-gender children, consistent with H1. The analysis will use a pooled sample from all available waves.

Estimation Strategy

- We will estimate a single ordered probit model on the pooled data.
- Standard errors will be clustered at the individual respondent level (`pid`) to account for repeated observations.
- The model will include the same controls as H1, plus year fixed effects (`i.syear`).

Interpretation

We consider the hypothesis supported if the coefficient on `has_only_daughters` is negative and statistically significant at the 5% level.

Hypothesis 3: Having daughters leads to more gender equality in the household (as measured by behaviors)

Dependent Variable

To measure behavior within the household, we will use the following single item:

- `pli0044_h`: A continuous measure of the hours spent on child care on an average weekday.

Main Independent Variable

The main independent variable is `has_only_daughters`, consistent with our overall strategy.

Sample Restrictions

- Adults aged 18 and above.
- Respondents with at least one child.
- The sample is restricted to respondents who are married or in a registered partnership and living together (derived from `pgfamstd`).

Estimation Strategy

- We will estimate a single Ordinary Least Squares (OLS) regression on the pooled data.
- Standard errors will be clustered at the individual respondent level (`pid`).
- The model will include the same controls as H2 (including year fixed effects).

Interpretation

We consider the hypothesis supported if the coefficient on `has_only_daughters` is positive and statistically significant at the 5% level.

Hypothesis 4: The effect of having daughters on support for gender equality is larger for men than for women

Dependent Variable

The dependent variable is the same as in H1:

- `plh0395i05`: The ordinal measure of attitudes toward non-traditional gender roles for children (2019 wave).

Main Specification

The analysis will be restricted to parents with only same-gender children. To test for the differential effect, we will estimate a model including an interaction term:

- The model will include `has_only_daughters`, a binary indicator for the respondent's gender (`male`), and the interaction between these two variables (`i.has_only_daughters##i.male`).

Estimation Strategy

- We will estimate a single OLS regression, treating the ordinal dependent variable as continuous for ease of interpreting the interaction term.
- The model will use robust standard errors.
- Control variables will be the same as in H1.

Interpretation

The focus is on the coefficient of the interaction term (`1.has_only_daughters#1.male`). We consider the hypothesis supported if this coefficient is positive and statistically significant at the 5% level.